

Empowering Education Through Ai-supported E-learning

R26-SE-041

Our Team



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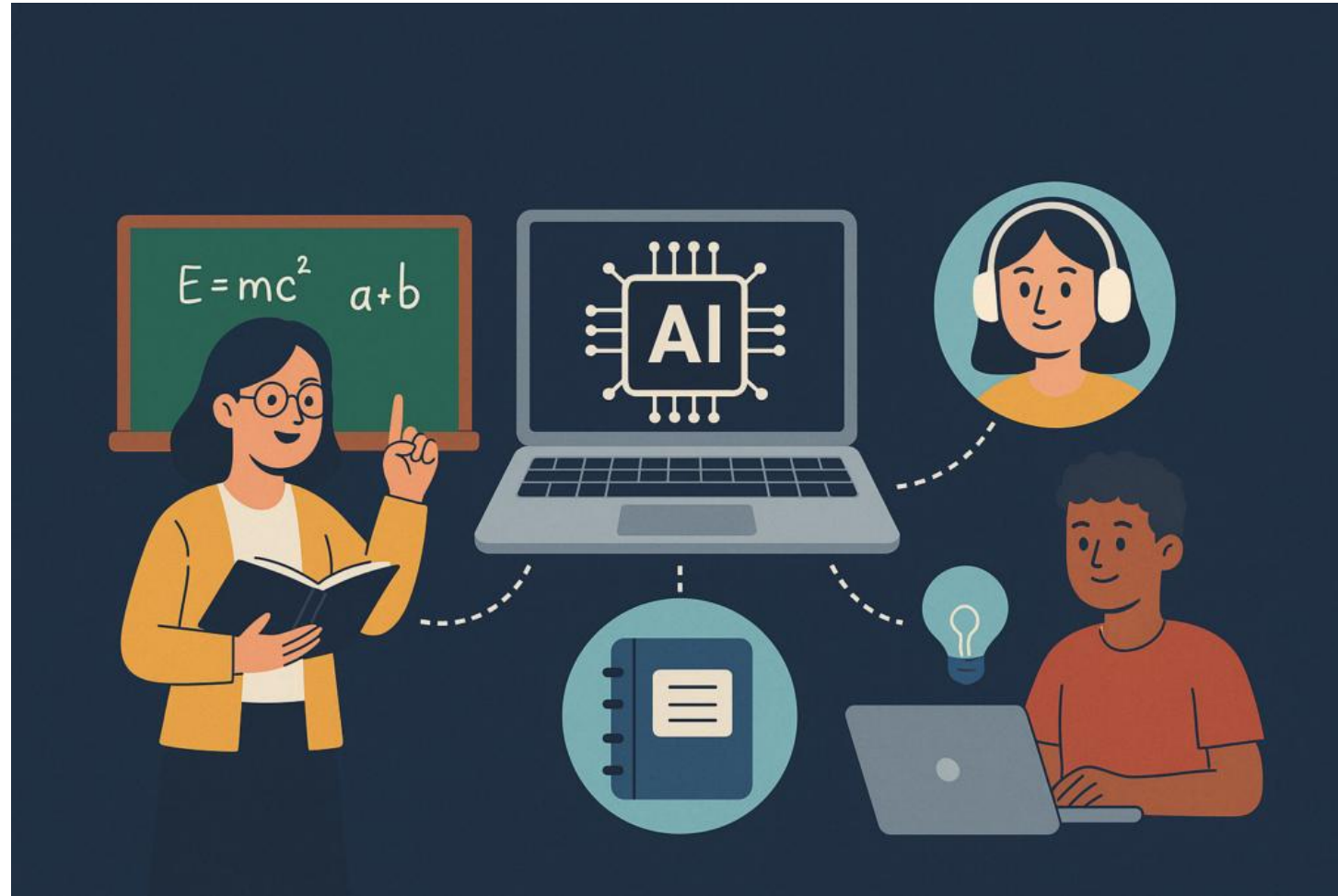


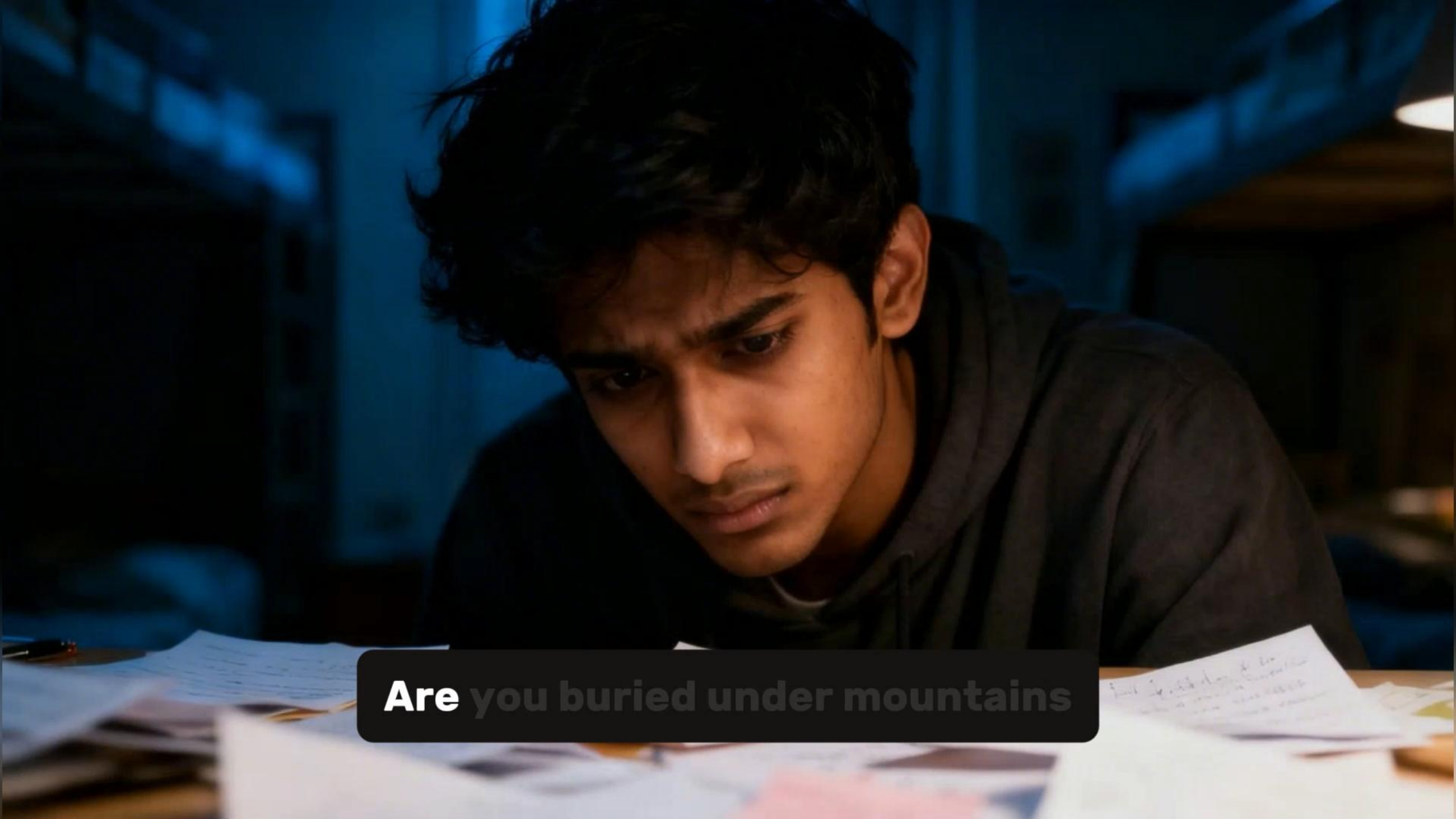
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Empowering secondary education through AI-Supported E-Learning





Are you buried under mountains

Empowering education through AI-Supported E-Learning

The Problem

- The standard "one-size-fits-all" teaching style doesn't work for everyone.
- It causes students to lose interest and miss important concepts.

Users & Impact

- Who it affects:** University students, teachers, and the universities.
- Social Impact:** Students fall behind and struggle to learn on their own.
- Economic Impact:** Students aren't ready for jobs on time, which hurts university success rates.

Current Tools & Their Failures

- Existing Tools:** Google NotebookLM, ChatGPT, Khanmigo.
- The Issues:** They only do basic tasks and lack deep personalization.
- The Gap:** They do not understand local languages like Sinhala and Tamil.

Our Proposed Solution

- What it is:** An all-in-one, AI-powered smart learning platform.
- How it works:** It brings together 4 unique AI tools to track student progress and automatically adjust the learning content and speed.

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Interactive Infographic and Visual Narrative Generator.

Turns student text notes into interactive, visual data stories. Helps students avoid information overload by making complex topics easy to see and explore.

Novelty (What makes it new?):

Existing tools only make flat, unclickable, non-interactive images.

My tool lets users click, explore, and find the exact source of the information.

INTERACTIVE INFOGRAPHIC AND VISUAL NARRATIVE GENERATOR

- **Problem Definition:**

Existing: Tools like Google NotebookLM only generate static, flat summaries from notes.

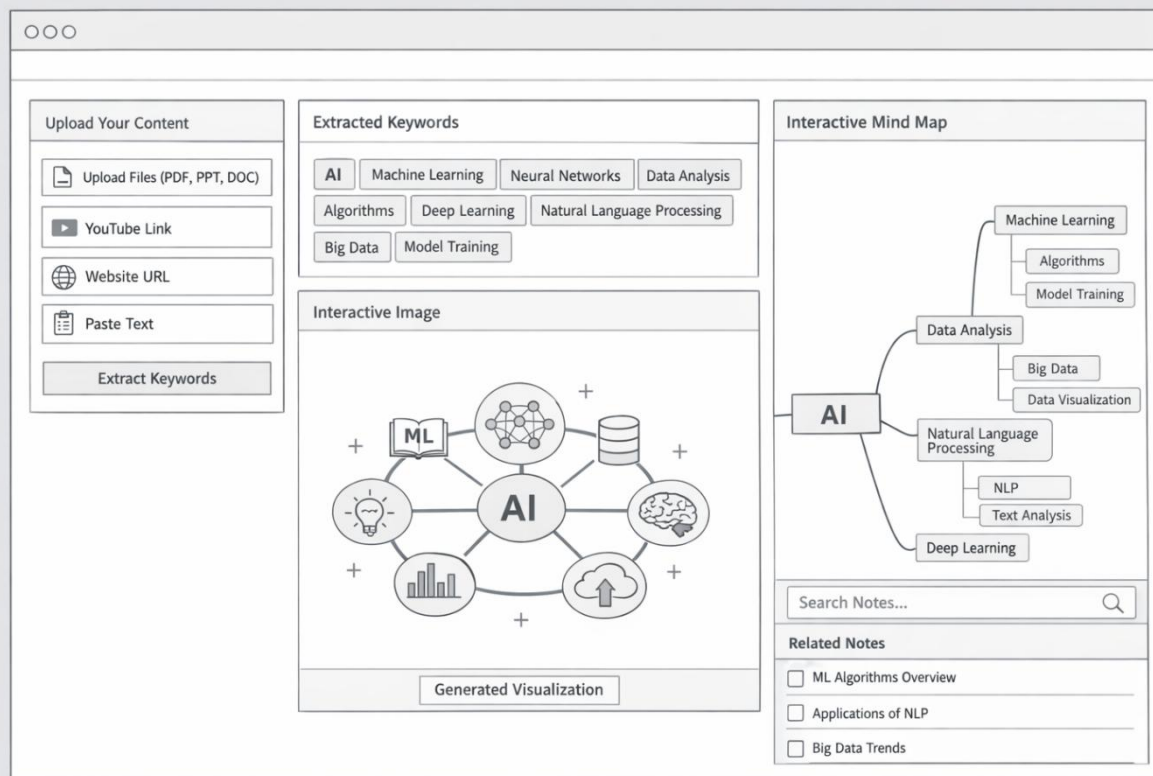
Gap: Students cannot click to explore the data or easily change how the chart looks.

Expectation: A system where students actively control and explore their visual study materials.

- **Research Objectives (SMART):**

1. Use open-source AI models to accurately pull important facts and relationships from text notes.
2. Build a smart AI feature that automatically suggests the best type of chart to use (like a timeline or a network).
3. Create a dynamic screen where users can filter information, click for details, and study together.

INTERACTIVE INFOGRAPHIC AND VISUAL NARRATIVE GENERATOR



Context Understanding Model
(Qwen2.5-VL / LLaVA-Next / DeepSeek-VL2 / InternVL2)



Scene & Prompt Planning



Image and mindmap Generation Model
(FLUX.1 / SDXL / Z-Image)



Interactive Vision Model (Region Grounding + Visual Q&A)
(Grounding DINO / Kosmos-2 / MiniGPT-4 / GPT4RoI)



User Interaction
(Circle area / Tap object / Ask question about region)

Interactive Infographic and Visual Narrative Generator

Application of Domain Knowledge & Methodology

- NLP & AI for extracting key entities, relationships, and statistics from user notes.
- Rule-based AI algorithms for automated chart type recommendation.
- Design Science Research (DSR) and experimental approaches.
- Interactive data visualization principles to reduce cognitive overload.

Validation & Evaluation

- Evaluate quantitative metrics: SLM extraction accuracy (precision/recall) and rendering latency.
- "Human-in-the-Loop" pilot study to assess UI usability and narrative control.
- Comparison of interactive visual models against traditional static models.

Technologies Used

- Hugging Face, spaCy, and NLTK (Open-source SLMs & NLP pipelines).
- Rule-based / Lightweight Machine Learning algorithms.
- Open-source front-end visualization charting libraries.
- Python (OOP) and Knowledge Graph databases.

Data & Ethics

- Controlled pilot study with volunteer undergraduates.
- Synthetic data (Gemini/Deepseek) & public datasets (**EdNet, ASSISTments, Kaggle, Huggingface**).
- Institutional Review Board (IRB) ethical clearance.
- Privacy ensured via data minimization and informed consent.

INTERACTIVE INFOGRAPHIC AND VISUAL NARRATIVE GENERATOR

System Overview (How it works):

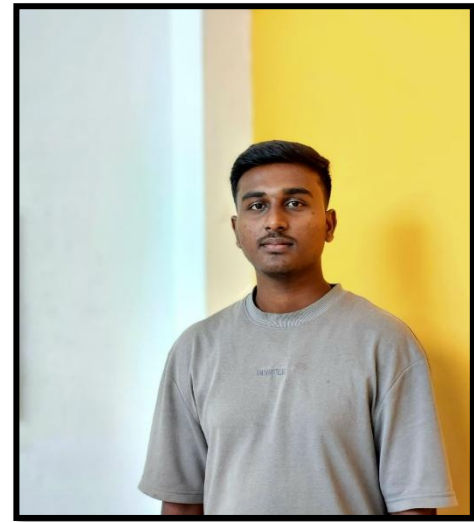
1. Input: User uploads text notes.
2. AI Extraction: System pulls out the main facts.
3. Smart Choice: AI picks the best interactive **Infograph**, **Mindmap** type.
4. Output: The interactive chart is shown to the user.

Dependencies and Common Components:

- My tool feeds into the main "Central Knowledge Orchestrator" shared by the whole team.
- The final charts will be displayed on the shared "Personalized Learning Interface" alongside my team's audio and quiz tools.

Interactive AI-Powered Audio Review

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Interactive AI-Powered Audio Review

Contribution Of Your Component To The Overall Solution

1. Changes passive listening into an active learning experience
2. Allows real-time voice Q&A directly with the audio
3. Supports local languages like Sinhala and Tamil

Novelty: Unlike existing tools, this system allows real-time voice interactions using local languages like Sinhala and Tamil



Interactive AI-Powered Audio Review

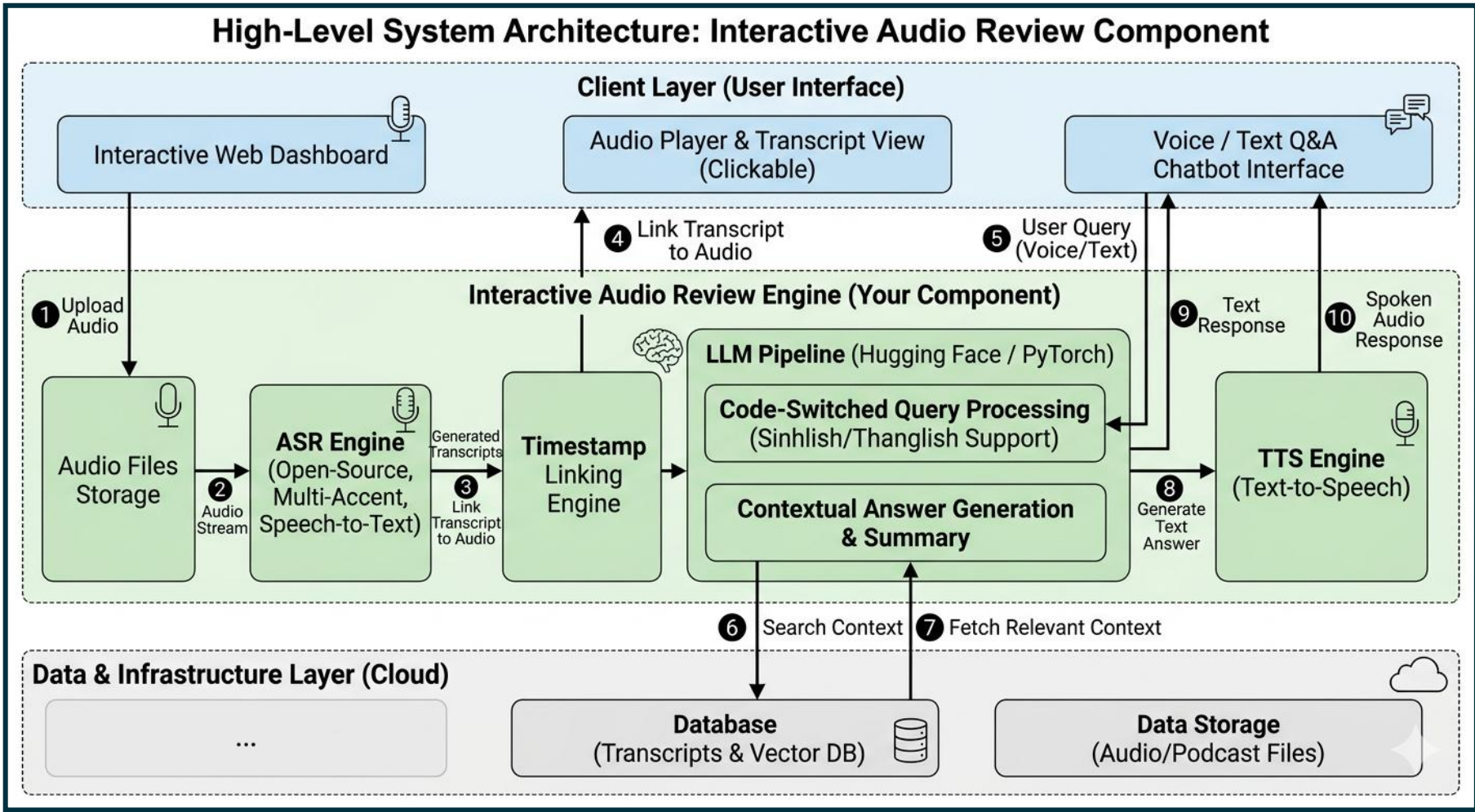
Problem Definition

- ✓ Passive audio learning (**e.g., recorded lectures, Notebook LM**) causes student disengagement because it is strictly linear
- ✓ Students lack the ability to dynamically query the audio stream or jump to specific concepts in real-time
- ✓ Existing proprietary platforms do not support mixed local languages (**Sinhala, Singlish and Tamil, Tanglish**), creating accessibility barriers

Research Objectives

- ✓ **Main:** To engineer an **interactive AI audio component** that transforms **passive listening into an active, conversational session**
- ✓ **Sub- objectives:**
 - > Transcribe educational audio using **open-source Automatic Speech Recognition (ASR)**
 - > Process transcripts and code-switched queries using an **open-source Large Language Model (SLM)**
 - > Enable real-time voice responses using integrated **Text-to-Speech (TTS) models**

Interactive AI-Powered Audio Review



Interactive AI-Powered Audio Review

Application of Domain Knowledge & Methodology

- ✓ Software Engineering (SE) and Artificial Intelligence (AI) domain application
- ✓ Design Science Research (DSR) approach for systematic creation
- ✓ Audio processing to transform passive listening into an active learning session

Validation & Evaluation

- ✓ Measuring transcription accuracy using Word Error Rate (WER)
- ✓ Testing system performance and response latency for real-time voice Q&A
- ✓ Evaluating user engagement and summary relevance during pilot testing

Technologies Used

- ✓ Open-source ASR, SLMs, and TTS models
- ✓ Hugging Face Transformers & PyTorch for model tuning
- ✓ Python, React.js, and Node.js for interface and backend
- ✓ AWS Cloud and MySQL/MongoDB for database and hosting

Data & Ethics

- ✓ Synthetic data generation (using SLMs) for Singlish/ Tanglish chat logs
- ✓ Controlled pilot study involving volunteer undergraduates
- ✓ Privacy ensured via data minimization, informed consent, and right to withdraw

Interactive AI-Powered Audio Review

Data Sets

✓ ASR Model (Speech-to-Text):

Data Source: Open datasets (like OpenSLR), Kaggle, Hugging face and YouTube tech lectures .

Required Size: 10 to 20 hours of audio with matching text per language.

✓ SLM Engine (Text Analysis & Q&A):

Data Source: AI-generated simulated student chats and course content (Synthetic Data).

Required Size: 1,000 to 3,000 Question & Answer pairs.

Interactive AI-Powered Audio Review

System Overview

- ✓ Uses an integrated open-source AI pipeline: ASR (for transcription), SLM (for text analysis), and TTS (for voice answers)
- ✓ **Clickable Transcripts:** Links the text directly to the audio, so users can click any word and instantly jump to that exact time.
- ✓ **Mixed-Language Chatbot:** Features a web dashboard where users can ask voice or text questions in Sinhala and Tamil.

Dependencies and Common Components

- ✓ Depends on the platform's Central Knowledge Orchestrator and unified Learner Model to personalize the interactions
- ✓ Shares the cloud database (MySQL/ MongoDB) to safely store the generated transcripts and user profiles
- ✓ Relies on the platform's common cloud infrastructure (like AWS) to store the uploaded educational audio and podcast files



Adaptive Quiz and Assessment Generation Engine

Contribution of your component to the overall solution:

Automatically generates multi-format questions (MCQs, short answers) from uploaded materials using NLP.

Adjusts quiz difficulty dynamically in real-time based on the student's evolving mastery level.

Provides context-aware Socratic hints and fetches targeted recommendation resources (via web scraping) instead of spoon-feeding direct answers.

Novelty of component:

Tight integration of a continuous Knowledge Tracing Model with dynamic question generation.

Quizzes are repurposed from static final tests into real-time diagnostic probes that directly update the learner's personal knowledge model, creating a closed learning loop.

Adaptive Quiz and Assessment Generation Engine

Problem Definition

Existing: Current platforms give the exact same static level quizzes to all students.

Gap: They do not adapt to individual learning paces and simply give direct answers when a student makes a mistake.

Expectation: A smart assessment system that adjusts difficulty in real-time and guides students to think independently.

Research Objectives

Main Objective: To develop an AI-powered engine using NLP and Knowledge Tracing to auto-generate quizzes and adjust difficulty based on real-time performance.

Sub-objectives:

- *Use NLP to automatically generate diverse questions (MCQs, short answers) from uploaded notes.*
- *Implement a Knowledge Tracing Model to track real-time concept mastery.*
- *Provide Socratic hints and fetch targeted external study resources for incorrect answers.*

Adaptive Quiz and Assessment Generation Engine

Application of Domain Knowledge & Methodology

- AI & Machine Learning for Knowledge Tracing.
- NLP for automatic question generation and Socratic hints.
- Design Science Research approach.
- Educational principles applied for adaptive learning paths.

Validation & Evaluation

- Evaluating the actual improvement in student understanding by comparing pre-test and post-test scores.
- Assessing the quality and correctness of the NLP-generated questions.
- Testing the response time to ensure the next adaptive question and web-scraped resources load with minimal delay.

Technologies Used

- Hugging Face (Open-source SLMs).
- Knowledge Tracing Algorithm.
- Retrieval-Augmented Generation (RAG) & Web Scraping.
- Python, React JS, MongoDB.

Data & Ethics

- Synthetic data (SLMs)
- Kaggle
- Hugging Face

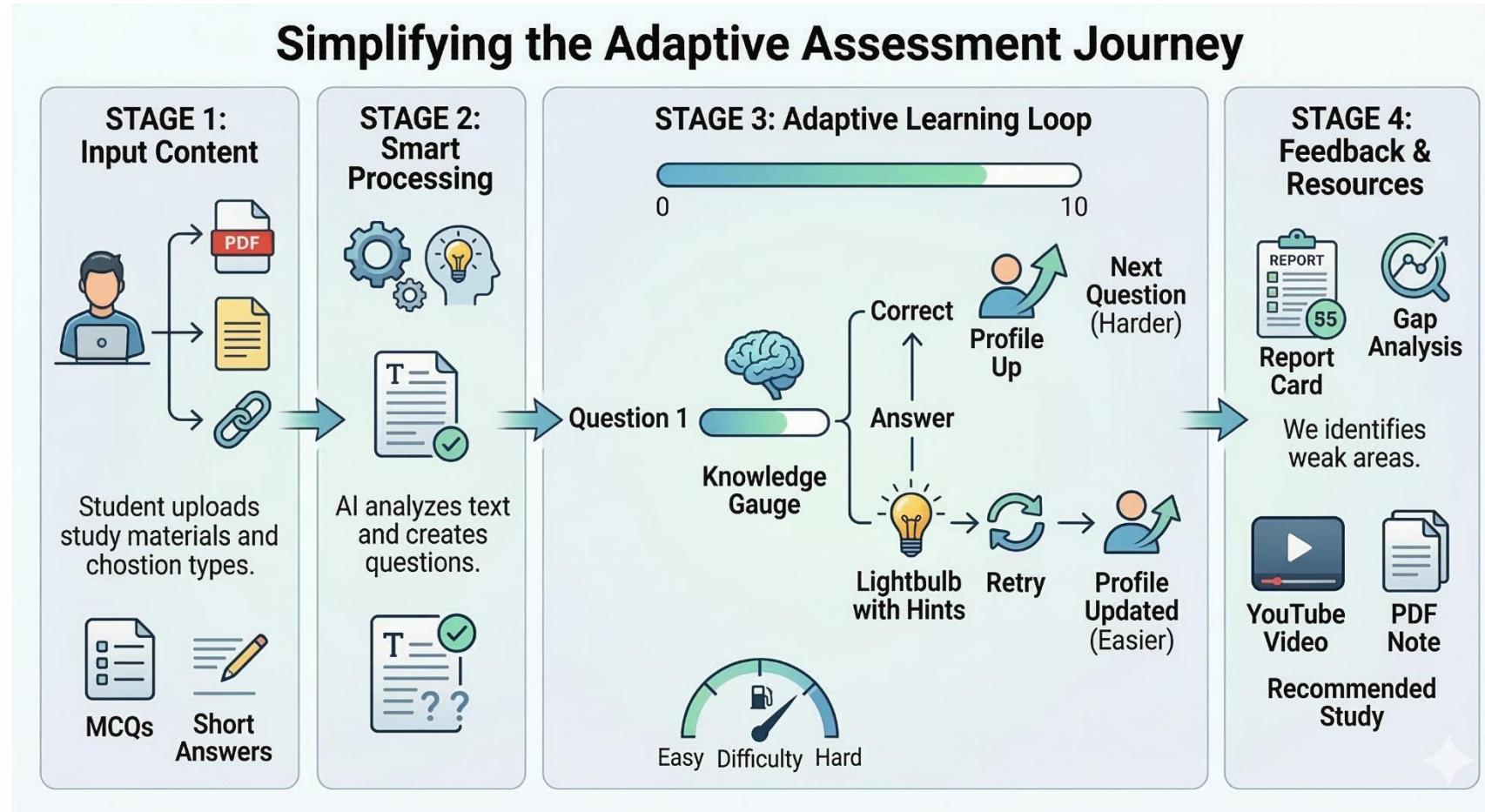
ADAPTIVE QUIZ AND ASSESSMENT GENERATION ENGINE

System Overview

- **Input:** Receives uploaded learning materials via the web interface.
- **Question Generation:** An NLP pipeline automatically generates diverse questions (MCQs, short answers).
- **Adaptation:** A Knowledge Tracing model evaluates answers in real-time and adjusts the difficulty of the next question.
- **Feedback System:** If an answer is wrong, it provides Socratic hints and fetches external resources (YouTube links) via web scraping.

ADAPTIVE QUIZ AND ASSESSMENT GENERATION ENGINE

SYSTEM OVERVIEW



ADAPTIVE QUIZ AND ASSESSMENT GENERATION ENGINE

Dependencies and Common Components

- Shared Learner Profile: Depends on the central database to read and update the student's overall mastery level.
- Central Knowledge Orchestrator: Syncs the quiz performance data with the platform's overall knowledge graph.
- Web Interface (UI): Relies on the common Student Dashboard to display the quizzes and hints to the user

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AI-Powered Handwritten Notes Extraction and Study Material Generation



Contribution of my component to the overall solution

- Extracts text from handwritten notes (Mainly Sinhala).
- Translates content from one language to another language.
- Transforms raw handwritten notes into structured, study-ready materials.
- Generates automatic flashcards.

AI-Powered Handwritten Notes Processing and Study Material Generation

Novelty (What makes it new?):

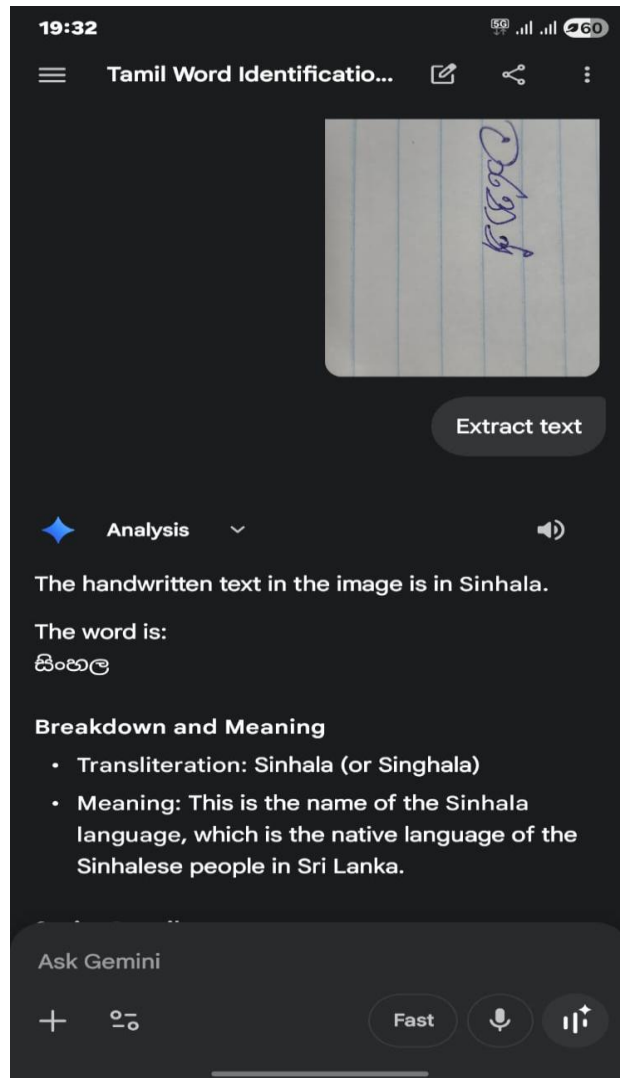
- Complete end-to-end pipeline: extraction, translation, structure study notes, flashcards in one system.
- Generate Structured Study notes. - Automatically organizes raw extracted text into structured study notes.
- Automatic flashcard generation.

PROBLEM DEFINITION

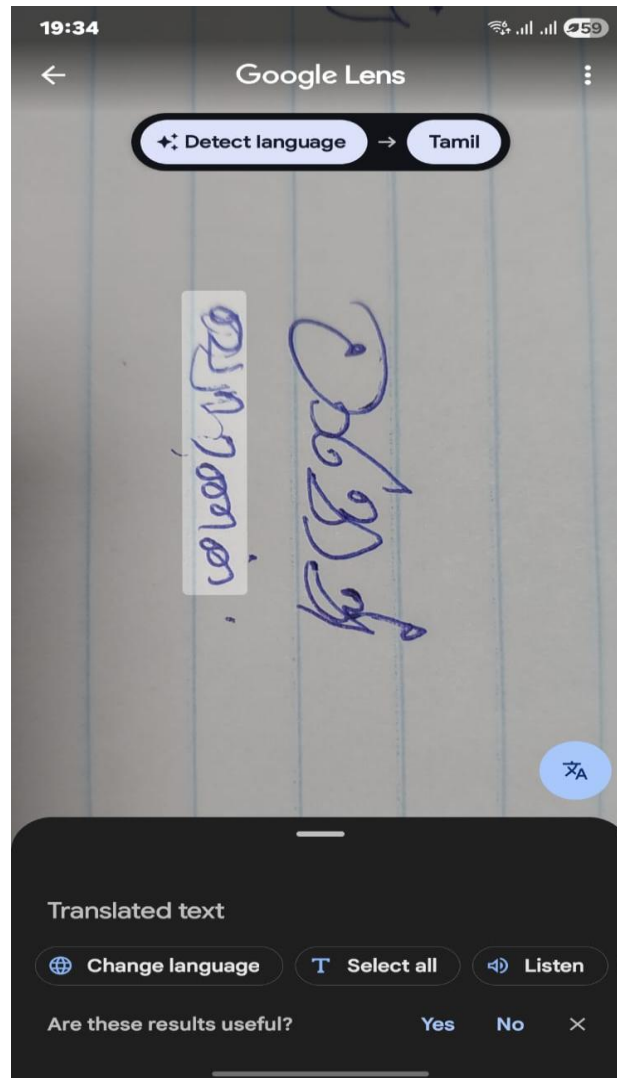
- **Existing tools** – Other Existing tools handle tasks separately and Often fail to accurately extract Sinhala handwritten text they only give unorganized raw text with no structure.
- **Research gap** - No single system combines Sinhala handwriting recognition, translation, note structuring and flashcard generation in one pipeline. Extracted text remains raw and unusable for studying.
- **My system** - Fills this gap by accurately recognizing Sinhala handwritten notes and automatically converting raw extracted text into structured study notes and flashcards in one complete pipeline.

Evidence of Existing AI Tools Failing to Extract Sinhala Handwritten Text

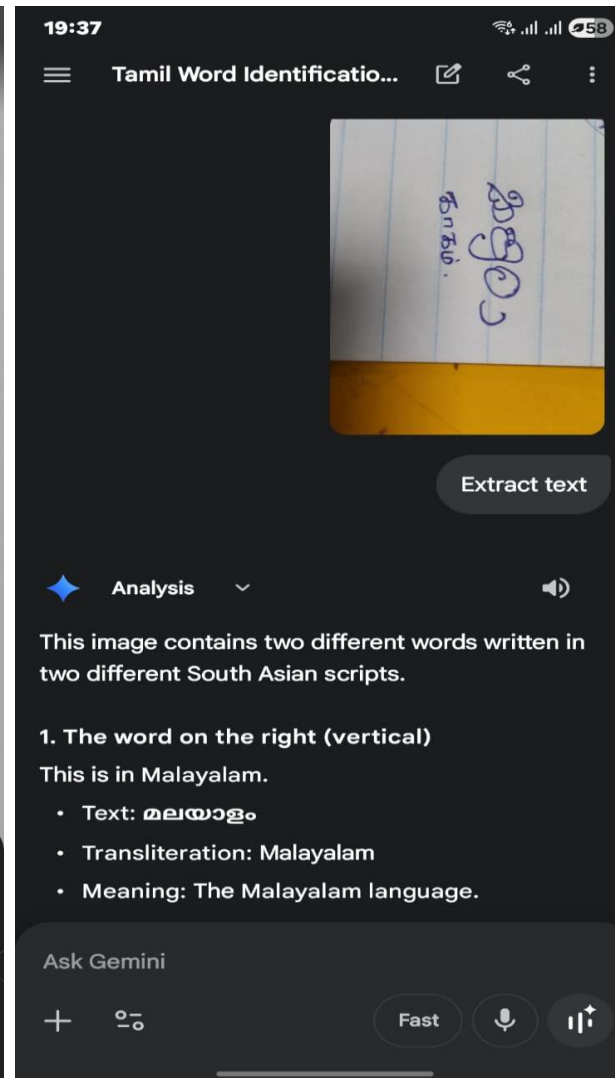
Gemini



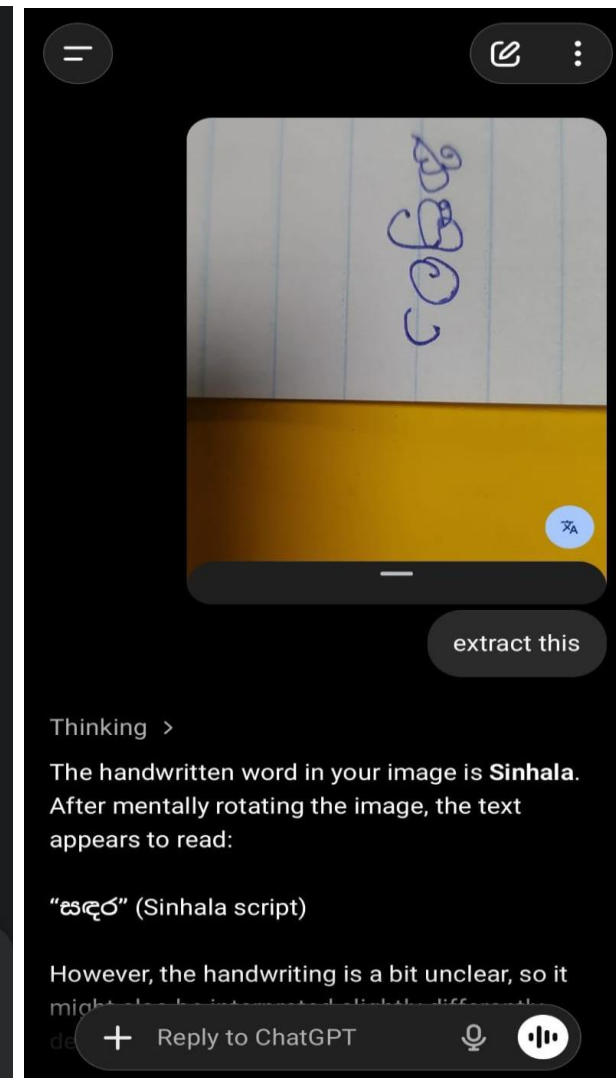
Google Lens



Gemini



ChatGPT



Main Objective:

- Develop an AI-powered system to Extract handwritten notes correctly and convert into structured, study-ready materials with flashcards

Sub-Objectives:

- Extract handwritten text notes of Sinhala accurately
- Translate notes seamlessly from Sinhala to Tamil, and English
- Structure notes by topics, headings, and contents
- Generate flashcards

AI-Powered Handwritten Notes Processing and Study Material Generation

Methodology

1. Input Handwritten notes images.
2. Text Extraction
3. Translation
4. Study-ready notes
5. Flashcard generation
6. Output Structured study notes and flashcards delivered to student

Validation & Evaluation

- Accuracy of text extraction
- Translation quality

Technologies Used

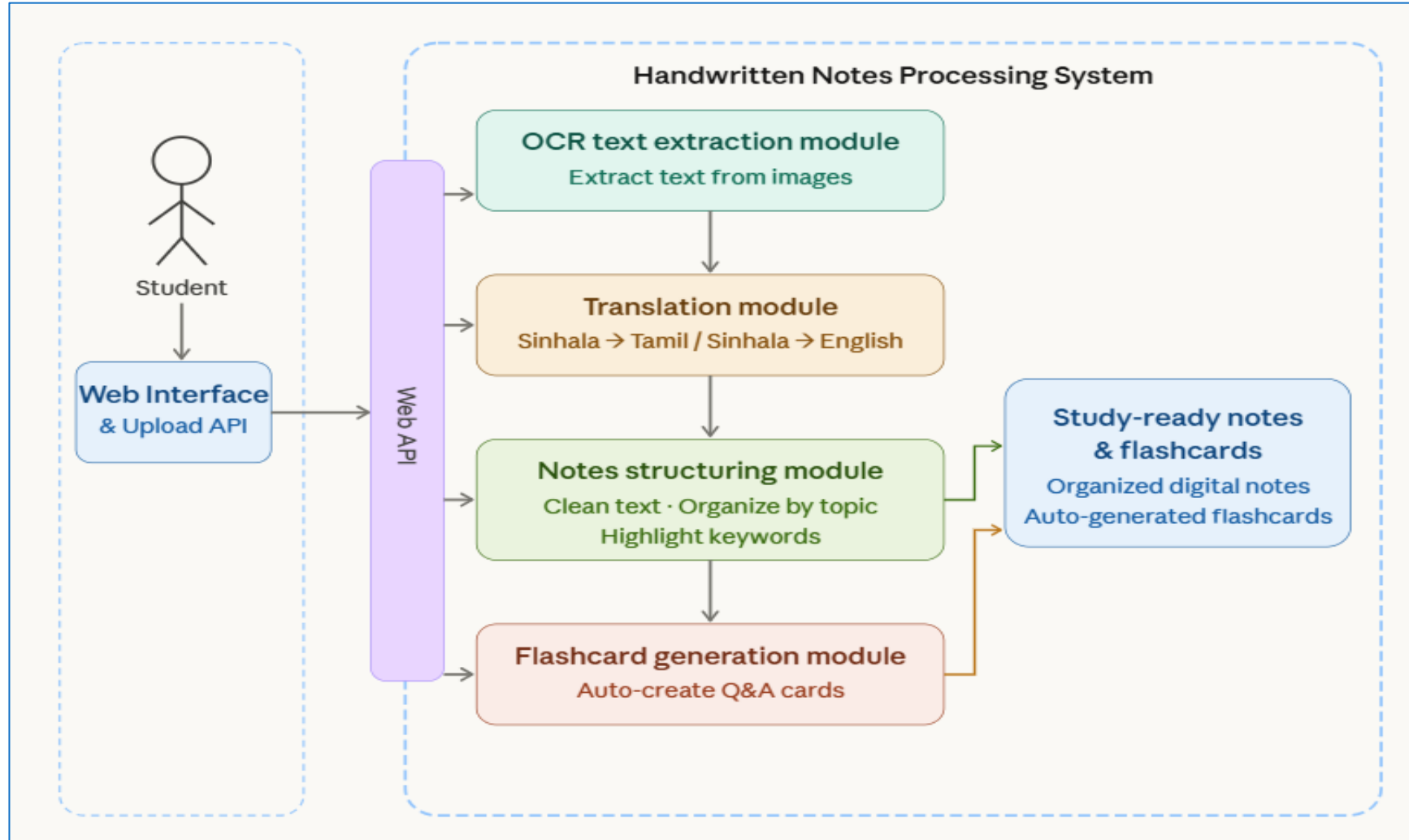
- Python, React.js, MongoDB
- TrOCR - Extracts handwritten Sinhala text from uploaded note images
- NLLB-200 -Translates extracted Sinhala text to Tamil and English
- Flan-T5 - Structures extracted text into organized study-ready notes with topics and bullet points
- Flan-T5 + RAG - Generates accurate Q&A flashcards from structured notes

Data & Ethics

- Student notes from consenting sources or public datasets
- Hugging Face
- Kaggle

AI-Powered Handwritten Notes Processing and Study Material Generation

System Overview



Empowering education through AI-Supported E-Learning

- Target Market:** The EdTech sector, specifically focusing on Higher Education and Lifelong Learning.
- Customer Profile:** University undergraduates who struggle with cognitive overload , and educators requiring AI-assisted course authoring tools.

09. Commercialization Aspects



BUSINESS MODEL

Individual



RS. 1000/month

- ✓ Feature Updates
- ✓ Ongoing Technical Support

Family



RS. 2500/month

- ✓ Up to 3 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support

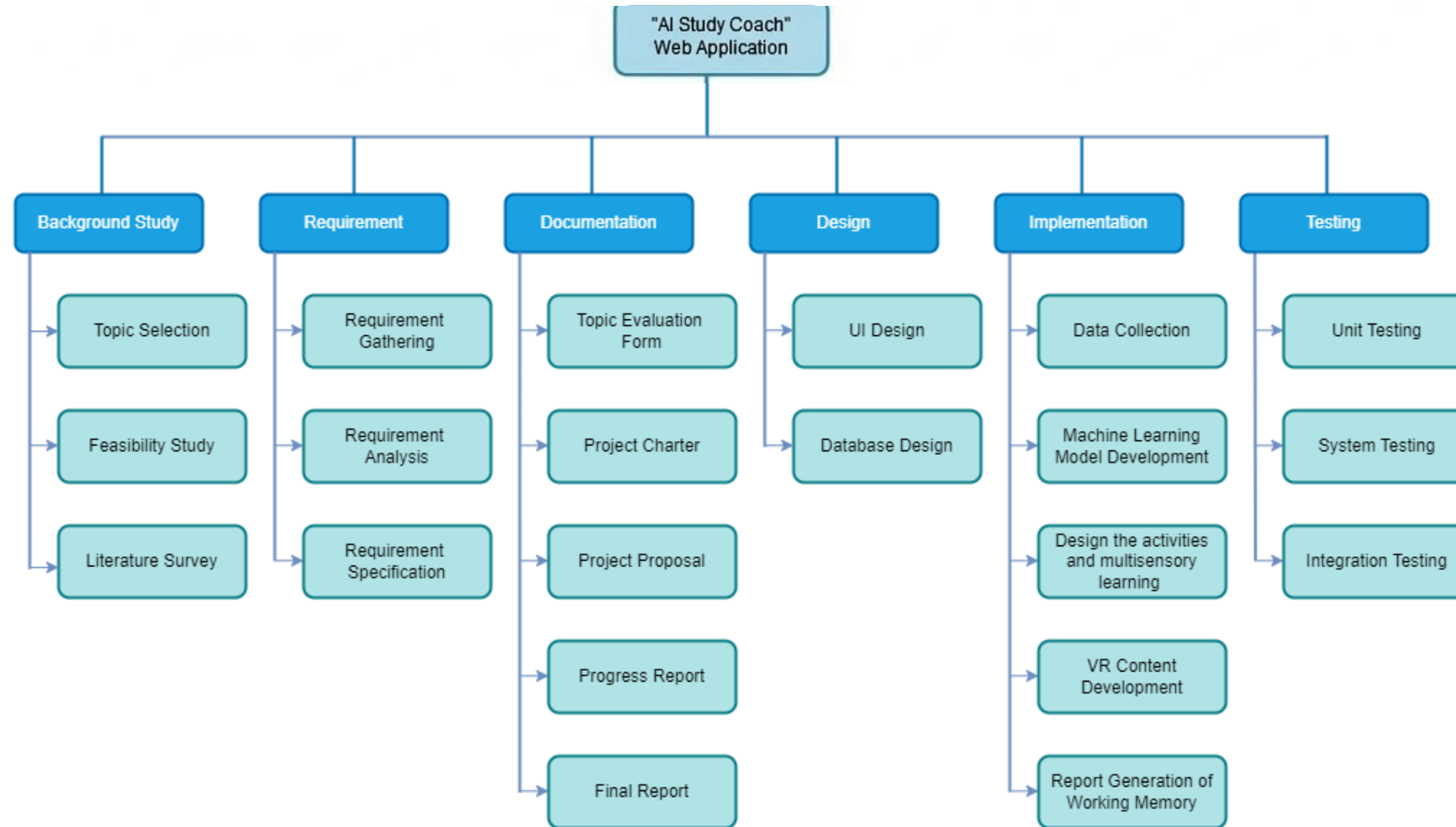
Classroom



RS. 5000/month

- ✓ Up to 40 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support

Work Breakdown Chart



Empowering education through AI-Supported E-Learning

PROJECT TIMELINE & GANTT CHART: RESEARCH PROPOSAL & REPORT (2026)

